

OSSP

Skill – 3 Codes

Skill3(i):

Code:

```
polisettyreelharsha@SREEHA x + v
GNU nano 8.7.1 Skill3code.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define MAX_HISTORY 10
#define MAX_CMD_LEN 100
char *history[MAX_HISTORY];
int history_count = 0;
void add_to_history(const char *cmd) {
    if (history_count < MAX_HISTORY) {
        history[history_count++] = strdup(cmd);
    } else {
        free(history[0]); // release oldest
        for (int i = 1; i < MAX_HISTORY; i++) history[i-1] = history[i];
        history[MAX_HISTORY-1] = strdup(cmd);
    }
}
void show_history() {
    for (int i = 0; i < history_count; i++) {
        printf("%d: %s\n", i+1, history[i]);
    }
}
int main() {
    char buffer[MAX_CMD_LEN];
    while (1) {
        printf("Enter command (or 'exit'): ");
        if (!fgets(buffer, sizeof(buffer), stdin)) break;
        buffer[strcspn(buffer, "\n")] = 0; // remove newline
        if (strcmp(buffer, "exit") == 0) break;
        add_to_history(buffer);
        show_history();
    }
    for (int i = 0; i < history_count; i++) free(history[i]);
    return 0;
}
```

Output:

```
polisettyreelharsha@SREEHA x + v
polisettyreelharsha@SREEHARSHA:~$ nano Skill3code.c
polisettyreelharsha@SREEHARSHA:~$ gcc Skill3code.c -o Skill3code
polisettyreelharsha@SREEHARSHA:~$ ./Skill3code
Enter command (or 'exit'): pwd
1: pwd
Enter command (or 'exit'): ls
1: pwd
2: ls
Enter command (or 'exit'): srjn
1: pwd
2: ls
3: srjn
Enter command (or 'exit'): exit
polisettyreelharsha@SREEHARSHA:~$ |
```

Skill 3(ii):

Code:

```
polisettyreearsha@SREEHA x + v
GNU nano 8.7.1 Skill4code.c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#define MAX_TOKENS 10
void parse_input(const char *input) {
    char *tokens[MAX_TOKENS];
    char *copy = strdup(input);
    int count = 0;
    char *tok = strtok(copy, " \\t\\n");
    while (tok && count < MAX_TOKENS) {
        tokens[count++] = tok;
        tok = strtok(NULL, " \\t\\n");
    }

    printf("Tokens:\\n");
    for (int i = 0; i < count; i++) {
        printf("[%d] %s\\n", i+1, tokens[i]);
    }

    free(copy);
}
int main() {
    char buffer[100];
    while (1) {
        printf("Enter command (or 'exit'): ");
        if (!fgets(buffer, sizeof(buffer), stdin)) break;
        buffer[strcspn(buffer, "\\n")] = 0;
        if (strcmp(buffer, "exit") == 0) break;
        parse_input(buffer);
    }
    return 0;
}
```

Output:

```
polisettyreearsha@SREEHA x + v
polisettyreearsha@SREEHARSHA:~$ nano Skill4code.c
polisettyreearsha@SREEHARSHA:~$ gcc Skill4code.c -o Skill4code.c
gcc: fatal error: input file 'Skill4code.c' is the same as output file
compilation terminated.
polisettyreearsha@SREEHARSHA:~$ gcc Skill4code.c -o Skill4code
polisettyreearsha@SREEHARSHA:~$ ./Skill4code
Enter command (or 'exit'): ls
Tokens:
[1] ls
Enter command (or 'exit'): pwd
Tokens:
[1] pwd
Enter command (or 'exit'): mkdir
Tokens:
[1] mkdir
Enter command (or 'exit'): exit
polisettyreearsha@SREEHARSHA:~$ |
```